Homework 12

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Contents

Problem 1

Let (X, d_X) and (Y, d_Y) be metric spaces, $f : X \to Y$, and $x_0 \in X$. Prove that f is continuous at x_0 if and only if for any open set $V \subseteq Y$ containing $f(x_0)$, there is an open set $U \subseteq X$ containing x_0 so that $x \in U$ implies $f(x) \in V$.

Discussion & Scratch Work

> This problem is asking me to prove that two definitions of continuity at a point are equivalent. From lecture, I know the standard ε–δ definition: f is continuous at x if for every $\varepsilon > 0$ there exists $\delta > 0$ such that
>
> $$d_X(x', x) < \delta \implies d_Y(f(x'), f(x)) < \varepsilon.$$
>
> I also know the open-set formulation: f is continuous at x if for every open set $V \subseteq Y$ with $f(x) \in V$, there exists an open set $U \subseteq X$ such that
>
> $$x \in U \qquad \text{and} \qquad f(U) \subseteq V.$$
>
> So the goal is to show these two definitions are equivalent. Since this is an "if and only if," I prove both directions.
>
> For the forward direction, I assume the ε–δ definition. If V is open and contains $f(x)$, then there exists an open ball $B_\varepsilon(f(x)) \subseteq V$. Then continuity gives a $\delta > 0$ such that $f(B_\delta(x)) \subseteq B_\varepsilon(f(x)) \subseteq V$. So I take $U = B_\delta(x)$.
>
> For the reverse direction, I assume the open-set condition. Given $\varepsilon > 0$, I consider the open ball $B_\varepsilon(f(x))$. By hypothesis, there exists an open set U containing x such that $f(U) \subseteq B_\varepsilon(f(x))$. Since U is open, there exists $\delta > 0$ such that $B_\delta(x) \subseteq U$. This gives the ε–δ condition.

Proof.
($\Rightarrow$) Assume that f is continuous at $x \in X$. Let $V \subseteq Y$ be open with $f(x) \in V$. Then there exists $\varepsilon > 0$ such that

$$B_\varepsilon(f(x)) \subseteq V.$$

By continuity at x, there exists $\delta > 0$ such that

$$d_X(x', x) < \delta \implies d_Y(f(x'), f(x)) < \varepsilon.$$

Let $U := B_\delta(x)$. Then U is open, $x \in U$, and if $x' \in U$, then $f(x') \in B_\varepsilon(f(x)) \subseteq V$. Hence

$$f(U) \subseteq V.$$

($\Leftarrow$) Assume that for every open set $V \subseteq Y$ with $f(x) \in V$, there exists an open set $U \subseteq X$ such that $x \in U$ and $f(U) \subseteq V$.

Let $\varepsilon > 0$ and set

$$V := B_\varepsilon(f(x)).$$

Then V is open and contains $f(x)$, so there exists an open set $U \subseteq X$ such that

$$x \in U \qquad \text{and} \qquad f(U) \subseteq B_\varepsilon(f(x)).$$

Since U is open and $x \in U$, there exists $\delta > 0$ such that

$$B_\delta(x) \subseteq U.$$

Now if $d_X(x', x) < \delta$, then $x' \in U$, so

$$f(x') \in B_\varepsilon(f(x)),$$

which implies

$$d_Y(f(x'), f(x)) < \varepsilon.$$

Therefore f is continuous at x. $\square$

Problem 2

Let (X, d_X) and (Y, d_Y) be metric spaces and $f : X \to Y$ a function. Prove that f is continuous if and only if $f(\overline{A}) \subseteq \overline{f(A)}$ for any $A \subseteq X$.

Discussion & Scratch Work

> This problem is asking me to prove an equivalent characterization of continuity using closures. I am given a function $f : X \to Y$ between metric spaces, and I need to show that
>
> $$f \text{ is continuous} \iff f(\overline{A}) \subseteq \overline{f(A)} \quad \text{for all } A \subseteq X.$$
>
> So I need to prove both directions.
>
> First, I unpack what this statement means. The set $\overline{A}$ consists of all points x that are either in A or are limits of sequences from A. Similarly, $\overline{f(A)}$ consists of all points in Y that are limits of sequences from $f(A)$.
>
> So the statement $f(\overline{A}) \subseteq \overline{f(A)}$ is saying: if x is either in A or is a limit of points in A, then $f(x)$ should be either in $f(A)$ or a limit of points in $f(A)$.
>
> This strongly suggests using the sequential characterization of closure and the fact from lecture that:
>
> $$f \text{ continuous} \iff x_n \to x \Rightarrow f(x_n) \to f(x).$$
>
> For the forward direction, I assume f is continuous. Then if $x \in \overline{A}$, I can find a sequence $a_n \in A$ with $a_n \to x$. Continuity will give $f(a_n) \to f(x)$, which means $f(x) \in \overline{f(A)}$.
>
> For the reverse direction, I should not use subsequences. Instead, I will use the open-set characterization from Problem 1. So I fix an open set $V \subseteq Y$ and a point $x \in X$ such that $f(x) \in V$, and I define
>
> $$A := X \setminus f^{-1}(V).$$
>
> If x were in $\overline{A}$, then the hypothesis would imply
>
> $$f(x) \in f(\overline{A}) \subseteq \overline{f(A)}.$$
>
> But every point of $f(A)$ lies in $Y \setminus V$, so $\overline{f(A)} \subseteq Y \setminus V$ because $Y \setminus V$ is closed. That would force $f(x) \notin V$, a contradiction. Hence $x \notin \overline{A}$.
>
> Since $x \notin \overline{A}$, there exists an open set U containing x with $U \cap A = \varnothing$. Therefore $U \subseteq X \setminus A = f^{-1}(V)$, so $f(U) \subseteq V$. This is exactly the open-set characterization of continuity.

Proof.
($\Rightarrow$) Assume that f is continuous. Let $A \subseteq X$, and let $x \in \overline{A}$. By the sequential characterization of closure, there exists a sequence $\{a_n\} \subseteq A$ such that

$$a_n \to x.$$

Since f is continuous, we have

$$f(a_n) \to f(x).$$

Each $f(a_n) \in f(A)$, so $f(x)$ is a limit of points in $f(A)$. Hence

$$f(x) \in \overline{f(A)}.$$

Therefore

$$f(\overline{A}) \subseteq \overline{f(A)}.$$

($\Leftarrow$) Assume that for all $A \subseteq X$, we have

$$f(\overline{A}) \subseteq \overline{f(A)}.$$

We show that f is continuous.

Let $V \subseteq Y$ be open, and let $x \in X$ satisfy

$$f(x) \in V.$$

Set

$$A := X \setminus f^{-1}(V).$$

We claim that $x \notin \overline{A}$. Suppose for contradiction that

$$x \in \overline{A}.$$

Then, since $x \in \overline{A}$, we have $f(x) \in f(\overline{A})$. By the hypothesis,

$$f(\overline{A}) \subseteq \overline{f(A)},$$

so

$$f(x) \in \overline{f(A)}.$$

Now if $y \in f(A)$, then $y = f(a)$ for some $a \in A$. Since $a \notin f^{-1}(V)$, we must have

$$f(a) \notin V.$$

Thus

$$f(A) \subseteq Y \setminus V.$$

Because V is open, the set $Y \setminus V$ is closed. Therefore

$$\overline{f(A)} \subseteq Y \setminus V.$$

It follows that

$$f(x) \in Y \setminus V,$$

which contradicts $f(x) \in V$.

Hence $x \notin \overline{A}$. Since $\overline{A}$ is closed, its complement is open, so there exists an open set $U \subseteq X$ such that

$$x \in U \qquad \text{and} \qquad U \cap A = \varnothing.$$

Therefore

$$U \subseteq X \setminus A = f^{-1}(V).$$

So if $u \in U$, then $u \in f^{-1}(V)$, which means $f(u) \in V$. Hence

$$f(U) \subseteq V.$$

We have shown that for every open set $V \subseteq Y$ containing $f(x)$, there exists an open set $U \subseteq X$ containing x such that $f(U) \subseteq V$. By Problem 1, this means that f is continuous at x. Since $x \in X$ was arbitrary, f is continuous on X. $\square$

Problem 3

Let $f : \mathbb{R} \to \mathbb{R}$ be the function $f(x) = \frac{1}{1+x^2}$.

(a) Prove that f is continuous.

(b) Find an open set $U \subseteq \mathbb{R}$ so that $f(U)$ is not open.

(c) Find a closed set $F \subseteq \mathbb{R}$ so that $f(F)$ is not closed.

(d) Find a set $A \subseteq \mathbb{R}$ so that $f(\overline{A}) \neq \overline{f(A)}$.

Discussion & Scratch Work

This problem is asking me to analyze a specific function

$$f(x) = \frac{1}{1+x^2}$$

from several perspectives: continuity, behavior of images of open/closed sets, and how closure interacts with the function.

For part (a), I need to prove f is continuous. From Lecture 30, I know I can use the sequential definition: f is continuous if $x_n \to x$ implies $f(x_n) \to f(x)$. This is the easiest approach here since rational functions behave well under limits.

For part (b), I need to find an open set U such that $f(U)$ is not open. So I need to recall that a set is not open if it fails to contain an open ball around some point. Intuitively, f maps $\mathbb{R}$ into $(0, 1]$, and the maximum value 1 is only achieved at $x = 0$. That suggests that images of open sets containing 0 will include 1 but not values larger than 1, which creates a boundary point—so the image won't be open.

For part (c), I need a closed set F such that $f(F)$ is not closed. From Lecture 31, continuous functions do not necessarily preserve closed sets unless the domain is compact. So I should look for a closed set whose image "misses" a limit point.

For part (d), I need to find a set A where

$$f(\overline{A}) \neq \overline{f(A)}.$$

From Problem 2, I know that continuity gives only one inclusion:

$$f(\overline{A}) \subseteq \overline{f(A)}.$$

So I need an example where this inclusion is strict. This typically happens when limit points are "collapsed" under the function, meaning some points in $\overline{f(A)}$ are not hit by $f(\overline{A})$.

Proof.

(a) Let $\{x_n\}$ be a sequence in $\mathbb{R}$ such that $x_n \to x$. Then We use the standard limit laws from earlier lectures: polynomials and reciprocal functions are continuous where defined. Then

$$x_n^2 \to x^2,$$

so

$$1 + x_n^2 \to 1 + x^2.$$

Since the function $t \mapsto \frac{1}{t}$ is continuous for $t \neq 0$, we have

$$\frac{1}{1 + x_n^2} \to \frac{1}{1 + x^2}.$$

Thus $f(x_n) \to f(x)$, so f is continuous.

(b) Let $U = \mathbb{R}$, which is open. Then

$$f(U) = (0, 1]\,.$$

This set is not open in $\mathbb{R}$ because $1 \in f(U)$, but there is no open interval around 1 contained in $(0, 1]$. Hence $f(U)$ is not open.

(c) Let $F = [0, \infty)$, which is closed in $\mathbb{R}$. Then

$$f(F) = (0, 1]\,.$$

As above, $(0, 1]$ is not closed because it does not contain 0, which is a limit point. Hence $f(F)$ is not closed.

(d) Let

$$A = (0, \infty).$$

Then

$$\overline{A} = [0, \infty).$$

So

$$f(\overline{A}) = f([0, \infty)) = (0, 1].$$

On the other hand,

$$f(A) = f((0, \infty)) = (0, 1),$$

so

$$\overline{f(A)} = [0, 1].$$

Thus

$$f(\overline{A}) = (0, 1] \neq [0, 1] = \overline{f(A)}.$$

$\square$

Problem 4

Consider the metric spaces $(\mathbb{R}, |\cdot|)$ and $(\mathbb{R}^2, d_2)$.

(a) Prove that the functions $f, g : \mathbb{R}^2 \to \mathbb{R}$ given by $f(x, y) = x$ and $g(x, y) = y$ are continuous. (Hint: Use sequences.)

(b) Prove that if $h : [0, 1] \to \mathbb{R}^2$ is a continuous function such that $h(0) = (x_0, y_0)$, $h(1) = (x_1, y_1)$, and $x_0 < 0 < x_1$, then there exists a point $t \in (0, 1)$ so that $h(t)$ lies on the y-axis.

Discussion & Scratch Work

For part (a), the problem explicitly tells me to use sequences, so I should use the sequential characterization of continuity from lecture: a function is continuous if whenever a sequence converges in the domain, its image converges in the codomain. Here the functions are just the coordinate projections

$$f(x, y) = x \qquad \text{and} \qquad g(x, y) = y.$$

So if $(x_n, y_n) \to (x, y)$ in $(\mathbb{R}^2, d_2)$, then by the definition of the Euclidean metric,

$$d_2((x_n, y_n), (x, y)) = \sqrt{(x_n - x)^2 + (y_n - y)^2} \to 0.$$

That immediately forces both $x_n \to x$ and $y_n \to y$, so the coordinate functions should be continuous.

For part (b), the key idea is to compose the given path $h : [0, 1] \to \mathbb{R}^2$ with the projection onto the first coordinate. If I define

$$\varphi(t) := f(h(t)),$$

then $\varphi : [0, 1] \to \mathbb{R}$ is continuous because h is continuous and part (a) says f is continuous. Also,

$$\varphi(0) = x_0 < 0 \qquad \text{and} \qquad \varphi(1) = x_1 > 0.$$

So now I can use the Intermediate Value Theorem from calculus on φ. That gives some $t \in (0, 1)$ with $\varphi(t) = 0$, which means the first coordinate of $h(t)$ is 0. In other words, $h(t)$ lies on the y-axis.

Proof.

(a) We first prove that $f : \mathbb{R}^2 \to \mathbb{R}$ given by $f(x, y) = x$ is continuous. Let $\{(x_n, y_n)\}$ be a sequence in $\mathbb{R}^2$ such that

$$(x_n, y_n) \to (x, y)$$

in $(\mathbb{R}^2, d_2)$. Then

$$d_2((x_n, y_n), (x, y)) = \sqrt{(x_n - x)^2 + (y_n - y)^2} \to 0.$$

Since
$$|x_n - x| \le \sqrt{(x_n - x)^2 + (y_n - y)^2} = d_2((x_n, y_n), (x, y)),$$
it follows by the squeeze theorem that
$$x_n \to x.$$
Hence
$$f(x_n, y_n) = x_n \to x = f(x, y).$$
Therefore f is continuous.

The proof for $g : \mathbb{R}^2 \to \mathbb{R}$ given by $g(x, y) = y$ is exactly the same. If $(x_n, y_n) \to (x, y)$, then
$$|y_n - y| \le \sqrt{(x_n - x)^2 + (y_n - y)^2} = d_2((x_n, y_n), (x, y)) \to 0,$$
so
$$y_n \to y.$$
Thus
$$g(x_n, y_n) = y_n \to y = g(x, y),$$
and therefore g is continuous.

(b) Let $h : [0, 1] \to \mathbb{R}^2$ be continuous, with
$$h(0) = (x_0, y_0), \qquad h(1) = (x_1, y_1), \qquad x_0 < 0 < x_1.$$
Define
$$\varphi : [0, 1] \to \mathbb{R}, \qquad \varphi(t) := f(h(t)),$$
where $f : \mathbb{R}^2 \to \mathbb{R}$ is the first-coordinate projection from part (a), namely $f(x, y) = x$.

Since h is continuous and f is continuous, the composition $\varphi = f \circ h$ is continuous on $[0, 1]$. Moreover,
$$\varphi(0) = f(h(0)) = f(x_0, y_0) = x_0 < 0,$$
and
$$\varphi(1) = f(h(1)) = f(x_1, y_1) = x_1 > 0.$$
By the Intermediate Value Theorem from calculus, which we may assume, there exists some $t \in (0, 1)$ such that
$$\varphi(t) = 0.$$
But $\varphi(t) = f(h(t))$ is exactly the first coordinate of $h(t)$. So the first coordinate of $h(t)$ is 0, which means
$$h(t) \in \{(0, y) : y \in \mathbb{R}\},$$
that is, $h(t)$ lies on the y-axis. $\square$

Problem 5

Going forward, you may assume (no proof necessary) any fact about $\sin x$ and $\cos x$ from calculus, such as $\sin, \cos : \mathbb{R} \to [-1, 1]$ are continuous functions. Prove that the function $f : \mathbb{R}^2 \to \mathbb{R}$ given by

$$f(x, y) = x^2 + y^2 + \sin(x - y)$$

has a global minimum. (Hint: Notice that we cannot directly apply the extreme value theorem here. However, we can apply it to any compact subset of $\mathbb{R}^2$.)

Discussion & Scratch Work

The goal is to prove that

$$f(x, y) = x^2 + y^2 + \sin(x - y)$$

has a global minimum on $\mathbb{R}^2$. The hint is important: I cannot apply the Extreme Value Theorem directly on all of $\mathbb{R}^2$, because $\mathbb{R}^2$ is not compact. So the idea is to first show that I only need to look on some compact subset of $\mathbb{R}^2$, and then apply the Extreme Value Theorem there.

First, I note that f is continuous. The functions $(x, y) \mapsto x^2 + y^2$ and $(x, y) \mapsto x - y$ are continuous, and by the problem statement I may assume that sin is continuous. Therefore $(x, y) \mapsto \sin(x - y)$ is continuous, and so f is continuous as a sum of continuous functions.

Next, I want a lower bound that tells me what happens far away from the origin. Since

$$-1 \le \sin(x - y) \le 1,$$

I get

$$f(x, y) \ge x^2 + y^2 - 1.$$

So if (x, y) is far from the origin, then $x^2 + y^2$ is large, and therefore $f(x, y)$ is large as well. This is the key coercive-type estimate.

To use this effectively, I want to compare that lower bound with the value of f at some specific point. A convenient choice is

$$\left(-\frac{1}{2}, \frac{1}{2}\right),$$

because then

$$x - y = -1,$$

so

$$f\left(-\frac{1}{2}, \frac{1}{2}\right) = \frac{1}{4} + \frac{1}{4} + \sin(-1) = \frac{1}{2} - \sin(1) < 0.$$

On the other hand, if $x^2 + y^2 \ge 1$, then

$$f(x, y) \ge x^2 + y^2 - 1 \ge 0.$$

So every point outside the closed unit disk has $f(x,y) \geq 0$, while the point $\left(-\frac{1}{2}, \frac{1}{2}\right)$ inside the closed unit disk gives a negative value. That means a global minimizer cannot occur outside the closed unit disk.

So I restrict to
$$K = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}.$$

This set is closed and bounded in $\mathbb{R}^2$, hence compact. Since f is continuous, the Extreme Value Theorem implies that f attains a minimum on K. Because every point outside K has value at least 0, while some point in K has value < 0, that minimum on K is automatically a global minimum on all of $\mathbb{R}^2$.

Proof.
Define
$$f : \mathbb{R}^2 \to \mathbb{R}, \qquad f(x,y) = x^2 + y^2 + \sin(x-y).$$

First, f is continuous on $\mathbb{R}^2$. Indeed, $(x,y) \mapsto x^2 + y^2$ is continuous, $(x,y) \mapsto x - y$ is continuous, and $\sin : \mathbb{R} \to \mathbb{R}$ is continuous, so $(x,y) \mapsto \sin(x-y)$ is continuous. Therefore f is continuous as a sum of continuous functions.

Next, since
$$\sin(x-y) \geq -1$$
for all $(x,y) \in \mathbb{R}^2$, we have
$$f(x,y) = x^2 + y^2 + \sin(x-y) \geq x^2 + y^2 - 1.$$

In particular, whenever $x^2 + y^2 \geq 1$, we get
$$f(x,y) \geq 0.$$

Now consider the point
$$p = \left(-\frac{1}{2}, \frac{1}{2}\right).$$

Then
$$f(p) = f\left(-\frac{1}{2}, \frac{1}{2}\right) = \frac{1}{4} + \frac{1}{4} + \sin(-1) = \frac{1}{2} - \sin(1).$$

Since $\sin(1) > \frac{1}{2}$, it follows that
$$f(p) < 0.$$

Also, p lies in the closed unit disk
$$K := \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}.$$

The set K is closed and bounded in $\mathbb{R}^2$, hence compact. Since f is continuous, the Extreme Value Theorem implies that there exists some $q \in K$ such that
$$f(q) \leq f(z) \qquad \text{for all } z \in K.$$

We claim that q is a global minimizer of f on $\mathbb{R}^2$. Let $z = (x, y) \in \mathbb{R}^2$.
If $z \in K$, then by definition of q,

$$f(q) \leq f(z).$$

If $z \notin K$, then $x^2 + y^2 > 1$, so from the estimate above,

$$f(z) \geq 0.$$

But

$$f(q) \leq f(p) < 0,$$

so again

$$f(q) < 0 \leq f(z).$$

Thus in every case,

$$f(q) \leq f(z) \qquad \text{for all } z \in \mathbb{R}^2.$$

Therefore f has a global minimum on $\mathbb{R}^2$. □

Problem 6

Let X and Y be metric spaces and $f : X \to Y$ a function. Prove that if X is compact and f is continuous and bijective, then the inverse function $f^{-1} : Y \to X$ is also continuous. (Hint: It suffices to show that the preimage of any closed subset of X under f^{-1} is closed in Y.)

Discussion & Scratch Work

The goal is to prove that $f^{-1} : Y \to X$ is continuous. The hint is telling me exactly which continuity criterion to use: it is enough to show that for every closed set $C \subseteq X$, the preimage of C under f^{-1} is closed in Y.

So I should start with an arbitrary closed set $C \subseteq X$ and understand what

$$(f^{-1})^{-1}(C)$$

actually is. Since f is bijective, this set is just

$$f(C).$$

So the real task is to prove that $f(C)$ is closed in Y whenever C is closed in X.

Now the compactness assumption is what makes this work. Since X is compact and C is closed in X, I know from lecture that C is compact. Since f is continuous, the image of a compact set under f is compact. And since Y is a metric space, compact subsets of Y are closed. So the chain is:

$$C \text{ closed in } X \implies C \text{ compact} \implies f(C) \text{ compact} \implies f(C) \text{ closed in } Y.$$

That is exactly what I need to show that f^{-1} is continuous.

Proof.
We prove that $f^{-1} : Y \to X$ is continuous by using the closed-set criterion.

Let $C \subseteq X$ be closed. Since X is compact and C is closed in X, it follows that C is compact.

Because $f : X \to Y$ is continuous, the image of a compact set under f is compact. Hence

$$f(C)$$

is compact in Y.

Now Y is a metric space, and in a metric space every compact set is closed. Therefore $f(C)$ is closed in Y.

Since f is bijective, we have

$$(f^{-1})^{-1}(C) = f(C).$$

Thus $(f^{-1})^{-1}(C)$ is closed in Y for every closed set $C \subseteq X$.

Therefore, by the closed-set criterion for continuity, f^{-1} is continuous. $\square$

Appendix: Toolbox